

Project #1: Printing Assignment (FFF & SLA)

MAE 4450 – Additive Manufacturing Process Physics, Materials, Properties, and Post Processing

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By:

Mattan Kelchner
Mae Sliwinski
Julian Stoller

Introduction and Process Details

1. Our self-identified level of expertise is expert.
 - a. Mae: 3D Printing for prototyping on Pursua and Bambus, has optimized settings for infill, supports, direction of print for nicer surface finish. SLA for research high tolerance accuracy for flame arrestor tube fitting and to prevent leakage.
 - b. Julian: 3D Printing for a project team, First Place Winner of the 2025 SME Digital Manufacturing Contest, used FDM machine for small batch manufacturing and rapid prototyping during biomedical device internship, owns a Bambu Lab printer
 - c. Mattan: 3D Printing for hobby and class, owns a Bambu printer.
2. We selected a [gear fidget toy](#) because we wanted to test a part that had a snap fit assembly, and complex features such as gear tolerancing to verify that parts interface and fit together to inform function. Additive Manufacturing is suitable for this because it is a one-time fabrication process in which we are verifying tolerances and fit. Additionally, we are using PLA. SLA is ideal for small parts with short-term use because the material properties degrade over time.

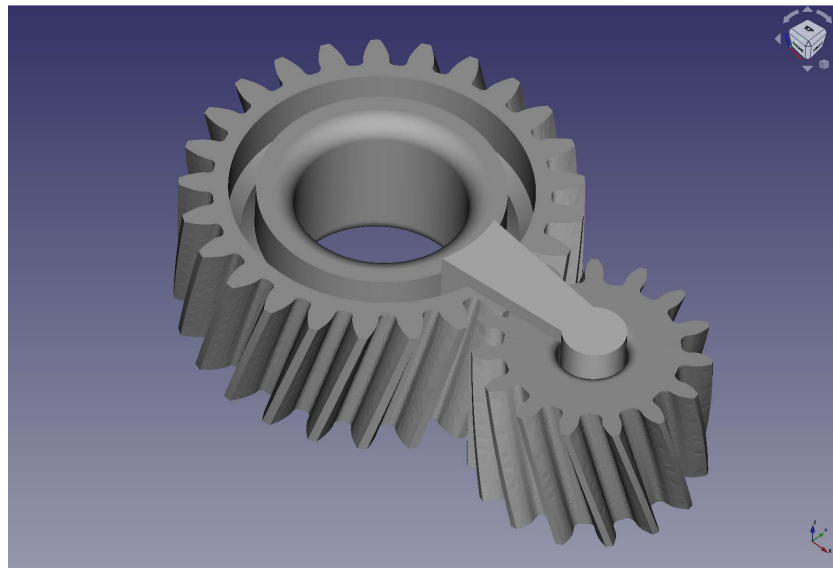


Figure 1: Photo of Assembled Model in CAD Software, Angled View

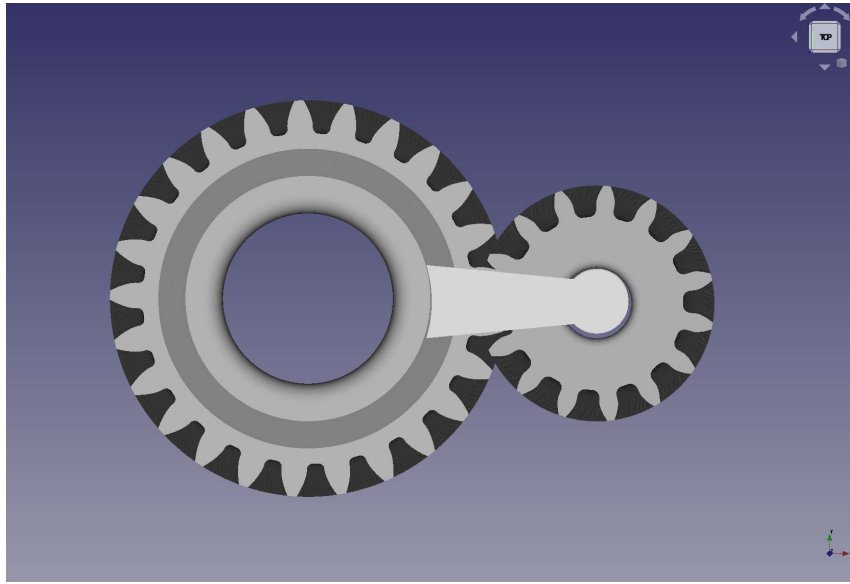


Figure 2: Photo of Assembled Model in CAD Software, Top View



Figure 3: Image of Gear-based Fidget Toy

3. Five key dimensions, considered quality:

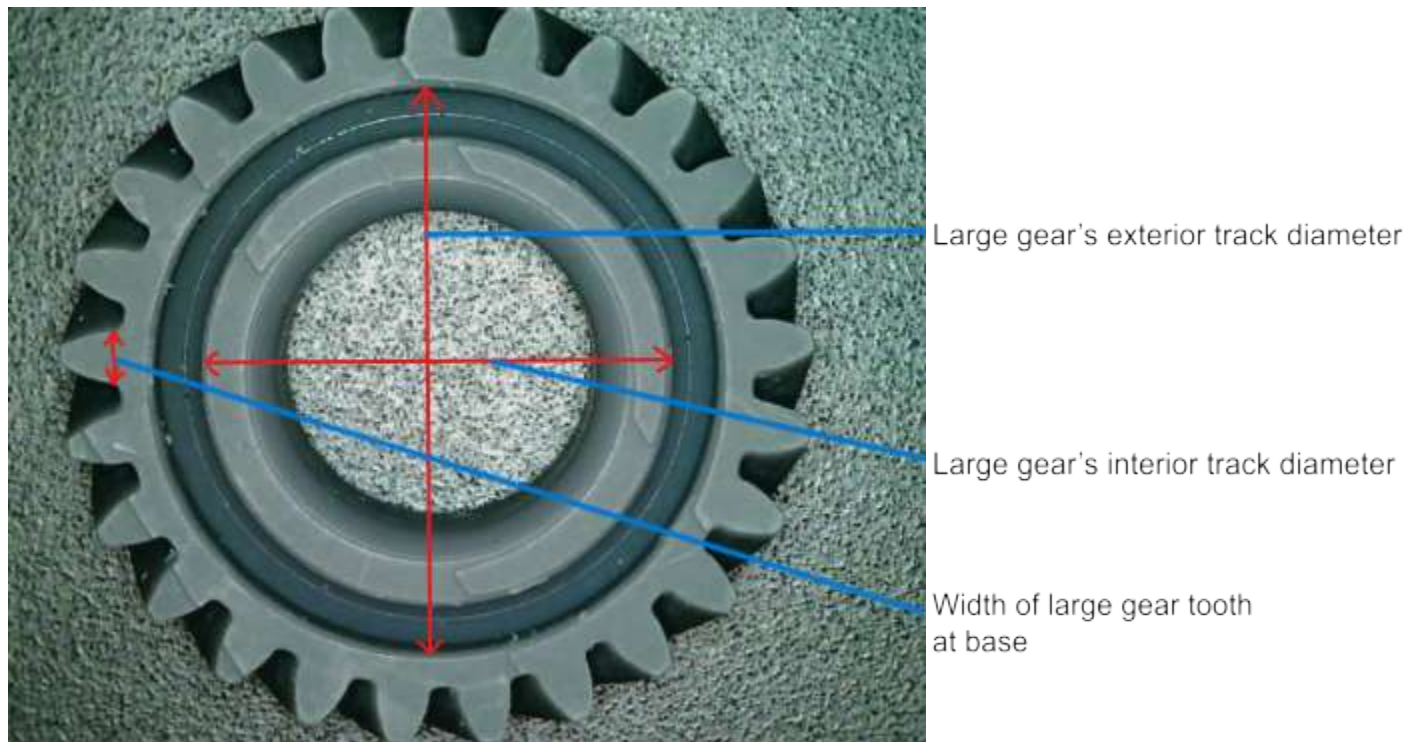


Figure 4: Large gear (dimensions analyzed for both SLA and PLA)

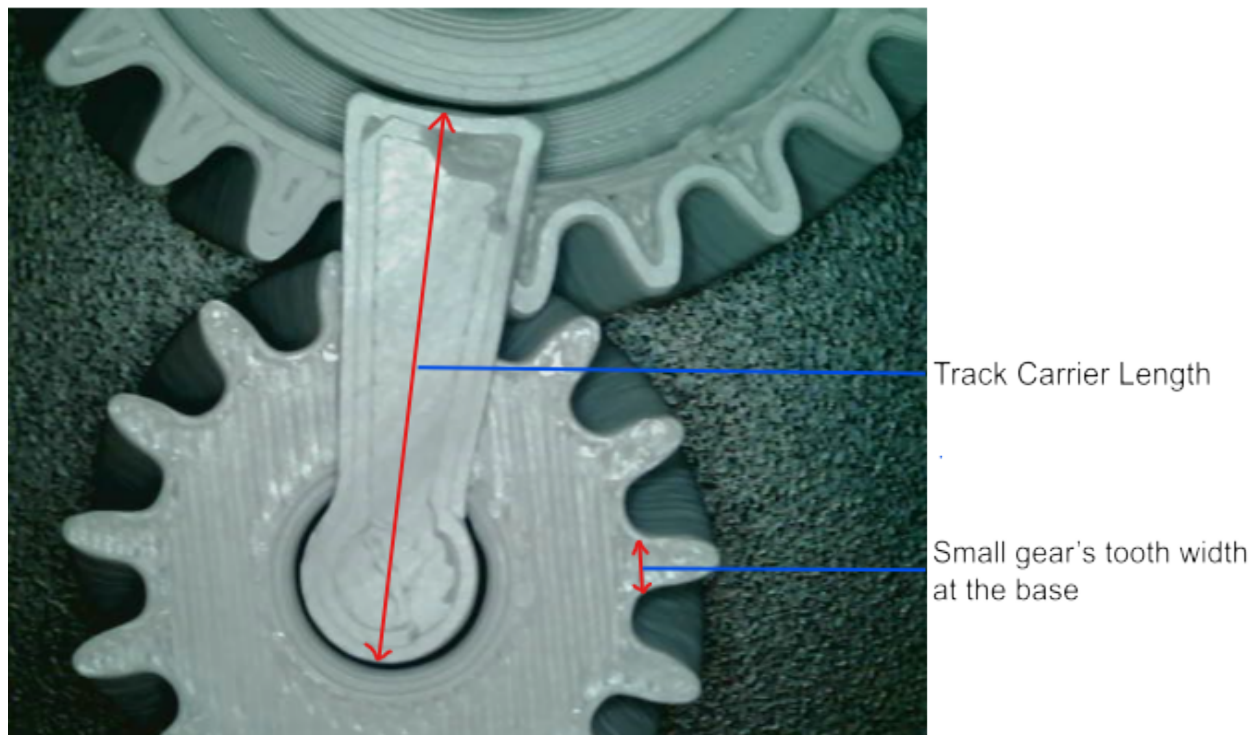


Figure 5: Small gear and track carrier (dimensions analyzed for both SLA and PLA)

Five key dimensions were selected as quality metrics because they directly influence the function and fit of the final assembly. The first two were the large gear track interior and exterior diameter, which reflect the overall size accuracy of the main part and indicates how well each process reproduces larger circular features. The large gear track interior diameter, additionally is important for evaluating concentricity and dimensional control of the internal ring geometry. The third and fourth were the tooth width at the base for both the large and small gears, since this feature directly affects tooth engagement, backlash, and rotational smoothness. Fifth, the track carrier length was measured because it determines the position of the small gear relative to the larger track and therefore affects assembly alignment. These dimensions were chosen to capture both global dimensional fidelity and the reproduction of smaller functional features.

4. Machines used, settings, and capabilities

The FFF parts were printed on a Bambu Lab X1C, an FDM printer equipped with a 0.4 mm nozzle and loaded with PLA filament. We used the printer's standard settings for this nozzle size: a 0.20 mm layer height, 15% sparse infill with a grid pattern, 2 wall loops, and no supports. The 0.20 mm layer height was considered appropriate for this part because the gears and sliding interfaces were not so small or delicate that a finer layer height was necessary to maintain functionality. Similarly, 15% infill and 2 wall loops were judged to be sufficient because the assembly was not expected to experience significant mechanical loading during either assembly or normal operation. For a lightly loaded part like this one, these settings provided a reasonable balance between structural adequacy and print efficiency.

The parts were printed without supports because the geometry of the assembly did not require them. Although the larger gear included a small bridged region, it was short enough that it could be printed successfully without additional support material. Avoiding supports also helped preserve the surface quality of the gear teeth and reduced post-processing effort. With these settings, the full FFF print took 47 minutes.

The SLA parts were printed on a Form 3 using Grey Pro V1 resin. A layer thickness of 0.050 mm was selected, while the remaining settings were kept at their defaults, including solid infill, auto-orient, and automatic support generation with fine-tip supports. In addition, each part was scaled down to 70% of its original size in order to reduce total print time. The fine 0.050 mm layer thickness was chosen because the two gears needed to mesh smoothly, and a smaller layer height helps reduce the surface texture that can increase friction along the gear teeth. A larger layer height would likely have produced a rougher tooth surface and made the meshing behavior less smooth.

Solid infill was used because the assembly was expected to experience some loading during handling and use, and a fully solid print provided greater confidence in the structural integrity of the part than a hollow print would have. Auto-orient was enabled to position the parts in a way that reduced the likelihood of residual cured resin collecting on important surfaces near the bed-facing regions. Because the resulting part orientations were angled, support material was required to stabilize the parts during the build. Fine-tip supports were selected because they were expected to leave smaller and less severe surface artifacts than other support options, which was preferable for preserving the quality of the gear teeth and other functional features. With these settings, the SLA print took 3 hours.

Qualitative Analysis

Defective Areas and Process Physics Root Cause

For our FFF print, on one side of the part that was attached to the bed there is some warping where you can see ridges and the stringy parts of the filament did not fully form a flat face. To prevent this in the future, we should use a raft to ensure proper bed adhesion to ensure the nozzle is hot enough before it prints the first layer and it would probably fill in the gaps leaving a smooth surface finish like the final layer once the nozzle is properly heated up. This is because the first layer has a tendency to want to delaminate from the build platform, and curl up to break adhesion when the gantry is moving fast, but the material has not reached the glass transition temperature for long enough when it optimally forms bonds. Another way to mitigate this issue would be to enable Bambu's feature of "Flow Dynamics Calibration" which is a firmware setting which improves 3D print quality by compensating for the lag in extrusion pressure. This is something that the RPL would have to enable when pressing send on our print, not something we can account for in the slicer.

Role of Build Orientation and Surface Finish Result

We chose to print the part flat on its smooth surface for FFF, with no supports to interfere with the gear teeth.

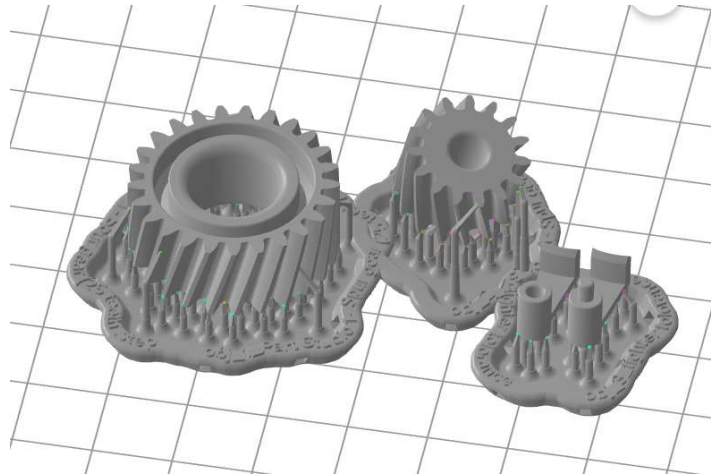


Figure 6: Image of Gear-based Fidget Toy in PreForm Slicer

For the SLA-printed part on the other hand, one side of the print has residual dots left over from the support structure that attaches it to the print bed. We chose to orient the small gear at a 45 degree angle to reduce the supports generated from interfering with the gear teeth, on the other hand the larger gear did not benefit from a 45 degree rotation so we left the part flat because the supports were fully on one end of the surface instead of attaching to the teeth. There were only a few anchor branches attaching to the gear teeth that did not leave noticeable impacts nor interfere with the function. This is an example of not always trusting the slicer to orient your parts, because we were able to visually check that an angle would not be optimal for this specific print due to the hollow ring structure in the larger gear. The attachment pieces to connect the larger gear to the smaller gear were printed with their flat end to build plate because orienting them at an angle interfered with the hole and peg; however, this might have been not beneficial to the structural integrity of the part as it was small and broke off due to the brittleness of removing the supports. In this case, SLA did not have proper tolerance of the hole and peg, and we had to force the two parts to connect using a blunt force whereas FFF succeeded on this front and connected properly. This demonstrates that SLA tends to shrink, whereas FFF tends to expand when compared to the dimensions from the CAD.

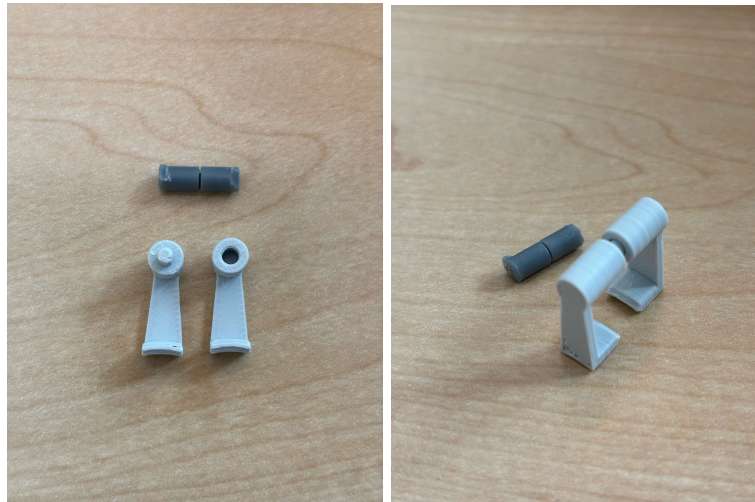


Figure 7: Peg and Hole (Dark Gray: SLA; Light Gray: FFF)

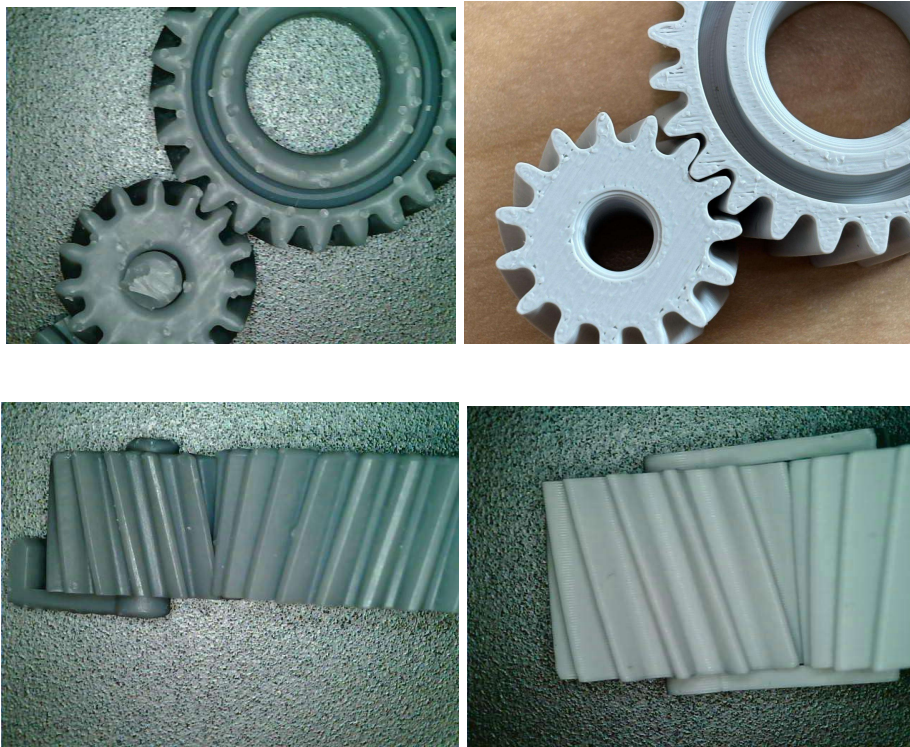
*** The arms broke off the SLA part when I tried to detach from assembly because I forced the parts together. The tolerance for the SLA part is a press fit tolerance, whereas the FFF part is a clearance tolerance that can rotate inside the hole, whereas the SLA peg does not budge once forced in.*

Comparison of Part Quality between AM Methods

The gear teeth on the SLA print are clearly defined, whereas on the FFF print the teeth do not have the flat defined sharp edge. The effect of corner rounding for FFF prints is stronger than SLA prints. The gears for the SLA print interlock at a tighter tolerance than the FFF print which does not entirely fill in the gap due to reduced structure in the teeth definition.



Figure 8: Image of Gear Teeth Definition (Dark Gray: SLA; Light Gray: FFF)



Influence of Post Processing

Post processing defects



Figure 9: SLA print after removal from print bed

Post-processing had a much larger effect on the SLA parts than on the FFF parts. For the SLA prints, washing and support removal were required before the parts could be inspected and assembled. These steps helped clean the parts and made the geometry easier to evaluate, but they also introduced some defects. In particular, the support contact points left small marks on the surface, and removing the supports likely weakened some of the thinner features, especially the connecting arms. This seems to have contributed to the damage that occurred during assembly and made the SLA parts less forgiving overall.

For the FFF print, post-processing had much less influence because the part was printed without supports and did not require the same amount of cleanup. Because of this, most of the defects seen on the FFF part were likely caused during printing rather than after it. The rough bottom surface appears to be more consistent with poor first-layer adhesion and slight curling on the build plate than with the actual geometry of the part. This is supported by the fact that the SLA version of the same geometry did not show the same defect on the corresponding surface.

Overall, post-processing improved the SLA part by making it usable and removing excess resin and supports, but it also introduced cosmetic defects and likely reduced the strength of some small features. The FFF part benefited from a simpler post-processing workflow, although its final quality was more dependent on the print itself, especially the first layer.

Quantitative Analysis

A set of digital calipers was used to measure the critical dimensions of the printed parts because the selected features were all accessible and could be measured with reasonable precision using a single tool.

The gear tooth width at the base was measured using the outside jaws of the calipers by aligning each jaw with opposite sides of the tooth base. Although this feature was relatively small, careful jaw placement allowed for reasonably consistent measurements. The large gear track interior diameter was measured using the inside jaws, with the calipers held parallel to the feature and expanded until both inner walls were contacted. The large gear track exterior diameter was measured using the outside jaws across the outer track surface. Finally, the track carrier length was measured as the linear

distance from the end of the circular section to the point where the carrier interfaces with the large gear track.

Calipers were chosen because they provided adequate precision for all selected dimensions while also allowing the measurements to be performed efficiently and repeatably.

Measurement Tables:

PLA Bambu Slicer Measurements

Feature	Measurement (mm)
Large gear tooth width (at base)	2.554
Small Gear tooth width (at base)	1.824
Large gear track interior diameter	$11.525 * 2 = 23.05$
Large gear track exterior diameter	$14.075 * 2 = 28.15$
Track Carrier Length	18.517

PLA Printed Parts

Feature	Measurement 1 (mm)	Measurement 2 (mm)	Measurement 3 (mm)	Measurement 4 (mm)	Measurement 5 (mm)
Large gear tooth width (at base)	2.87	2.82	2.91	2.88	2.81
Small Gear tooth width (at base)	2.22	2.27	2.27	2.21	2.22
Large gear track interior diameter	22.91	22.81	22.89	22.82	22.86
Large gear track exterior diameter	27.86	27.88	27.90	27.86	27.91

Track Carrier Length	18.76	18.76	18.76	18.76	18.76
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SLA Printed parts

Feature	<u>Measurement 1 (mm)</u>	<u>Measurement 2 (mm)</u>	<u>Measurement 3 (mm)</u>	<u>Measurement 4 (mm)</u>	<u>Measurement 5 (mm)</u>
Large gear tooth width (at base)	2.10	2.06	1.99	2.07	2.03
Small Gear tooth width (at base)	1.35	1.54	1.65	1.63	1.47
Large gear track interior diameter	16.33	16.28	16.23	16.30	16.24
Large gear track exterior diameter	19.63	19.58	19.57	19.66	19.53
Track Carrier Length	13.35	13.31	13.36	13.33	13.35

Important Note: Scaling was an issue and was caught after printing (byproduct of file formatting error before class announcement (we were the reason for the announcement))

Scaling factor applied to PLA to try bringing measurements into parity.

Parity PLA table (multiplying dimensions by .7, so scaling set to SLA scale)

Feature	<u>Measurement 1 (mm)</u>	<u>Measurement 2 (mm)</u>	<u>Measurement 3 (mm)</u>	<u>Measurement 4 (mm)</u>	<u>Measurement 5 (mm)</u>
Large gear tooth width (at base)	2.01	1.97	2.04	2.02	1.97

Small Gear tooth width (at base)	1.55	1.59	1.59	1.55	1.55
Large gear track interior diameter	16.04	15.97	16.02	15.97	16.00
Large gear track exterior diameter	19.50	19.52	19.53	19.50	19.54
Track Carrier Length	13.13	13.13	13.13	13.13	13.13

PLA printed part

- Large gear tooth width at base: average = 2.858 mm, variance = 0.00177 mm²
- Small gear tooth width at base: average = 2.238 mm, variance = 0.00087 mm²
- Large gear track interior diameter: average = 22.858 mm, variance = 0.00187 mm²
- Large gear track exterior diameter: average = 27.882 mm, variance = 0.00052 mm²
- Track carrier length: average = 18.760 mm, variance = 0.00000 mm²

SLA printed part

- Large gear tooth width at base: average = 2.050 mm, variance = 0.00175 mm²
- Small gear tooth width at base: average = 1.528 mm, variance = 0.01512 mm²
- Large gear track interior diameter: average = 16.276 mm, variance = 0.00173 mm²
- Large gear track exterior diameter: average = 19.594 mm, variance = 0.00263 mm²
- Track carrier length: average = 13.340 mm, variance = 0.00040 mm²

Parity-scaled PLA values

- Large gear tooth width at base: average = 2.002 mm, variance = 0.00097 mm²
- Small gear tooth width at base: average = 1.566 mm, variance = 0.00048 mm²
- Large gear track interior diameter: average = 16.000 mm, variance = 0.00095 mm²
- Large gear track exterior diameter: average = 19.518 mm, variance = 0.00032 mm²
- Track carrier length: average = 13.130 mm, variance = 0.00000 mm²

The repeated measurements show that both processes were fairly repeatable on larger circular features, since the track diameters had relatively small variances. However,

the smaller and more delicate features showed more process sensitivity. In particular, the SLA small-gear tooth dimension had the largest variance, which suggests that either the feature was more difficult to measure consistently or that the fine tooth geometry was more strongly affected by local curing, support interaction, or post-processing damage. In general, FFF produced dimensions that were slightly larger and more rounded, while SLA tended to produce sharper but sometimes undersized or tighter-fit features. This trend is consistent with the functional observations from assembly: the FFF “peg-hole” connection behaved more like a clearance fit, while the SLA version behaved more like an interference fit. Because a scaling issue occurred before printing, the direct comparison between the raw PLA and raw SLA dimensions should be interpreted cautiously. The parity-scaled PLA table is more useful for comparing overall dimensional trends than for claiming absolute process accuracy.

Direct comparison section:

PLA (FFF) average vs original expected table

Feature	Expected (mm)	PLA Avg (mm)	Percent Error
Large gear tooth width at base	2.554	2.858	+11.90%
Small gear tooth width at base	1.824	2.238	+22.70%
Large gear track interior diameter	23.050	22.858	-0.83%
Large gear track exterior diameter	28.150	27.882	-0.95%
Track carrier length	18.517	18.760	+1.31%

Scaled expected table (expected $\times$ 0.7) vs SLA average

Feature	Scaled Expected (mm)	SLA Avg (mm)	Percent Error
Large gear tooth width at base	1.788	2.050	+14.67%
Small gear tooth width at base	1.277	1.528	+19.67%
Large gear track interior diameter	16.135	16.276	+0.87%
Large gear track exterior diameter	19.705	19.594	-0.56%

Track carrier length	12.962	13.340	+2.92%
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PLA parity table vs SLA average

Feature	PLA Parity Avg (mm)	SLA Avg (mm)	Percent Difference
Large gear tooth width at base	2.002	2.050	2.37%
Small gear tooth width at base	1.566	1.528	2.46%
Large gear track interior diameter	16.000	16.276	1.71%
Large gear track exterior diameter	19.518	19.594	0.39%
Track carrier length	13.130	13.340	1.59%

After recognizing the scaling mismatch, a more meaningful comparison was made in three stages. First, the PLA (FFF) average dimensions were compared to the original expected dimensions from the slicer table. This showed that the larger features were reproduced fairly accurately, with errors of -0.83% and -0.95% for the large gear interior and exterior track diameters, and +1.31% for the track carrier length. However, the smaller tooth-base features were oversized by +11.90% and +22.70%, indicating that small local features were more strongly affected by the FFF process.

Next, the expected dimensions were scaled by a factor of 0.7 and compared against the SLA averages. Under this corrected comparison, the SLA print also showed good agreement on the larger features, with errors of +0.87% for the interior track diameter and -0.56% for the exterior track diameter, while the smaller tooth features remained more deviant at +14.67% and +19.67%.

Finally, the parity-scaled PLA values were compared directly to the SLA averages. This showed percent differences ranging from only 0.39% to 2.46%, indicating that once the dimensional scaling issue was accounted for, the PLA and SLA parts were actually (theoretically) much closer in size than the raw measurements initially suggested.

However, ultimately, we know the flaw in this comparison is that we're suffering from mis-scaled parts. This serves just to provide a rough approximation and recognize that this does not necessarily account for deviations that would occur when undergoing the process at a smaller scale in the case of the FFF printing.

Deviation justification

The measured deviations are consistent with the expected behavior of each process. In FFF, molten filament is deposited as finite-width roads, which tends to round corners, slightly enlarge some external features, and reduce accuracy on small detailed geometry such as the gear teeth. First-layer adhesion effects and thermal contraction during cooling can also contribute to dimensional error. In SLA, the dimensions are influenced by resin curing behavior, shrinkage during cure and post-cure, and local effects from support placement and removal. As a result, SLA can produce sharper visible features, but mating dimensions such as holes and pegs may become tighter than intended. These process-level effects are consistent with the measured deviations and with the observed difference in fit between the FFF and SLA parts.

Print Time Analysis

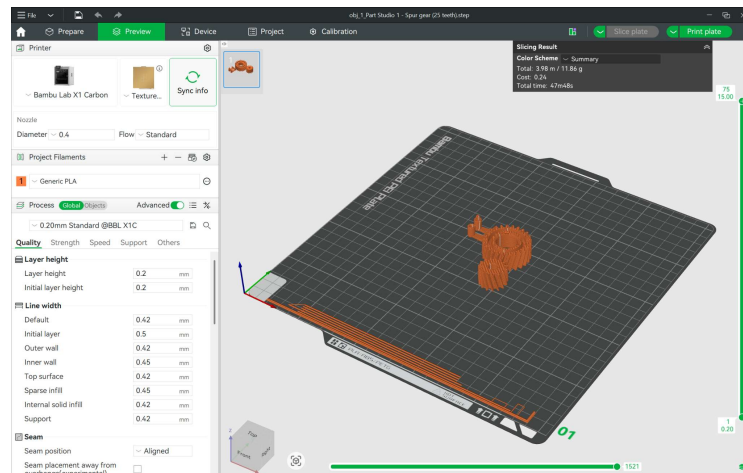


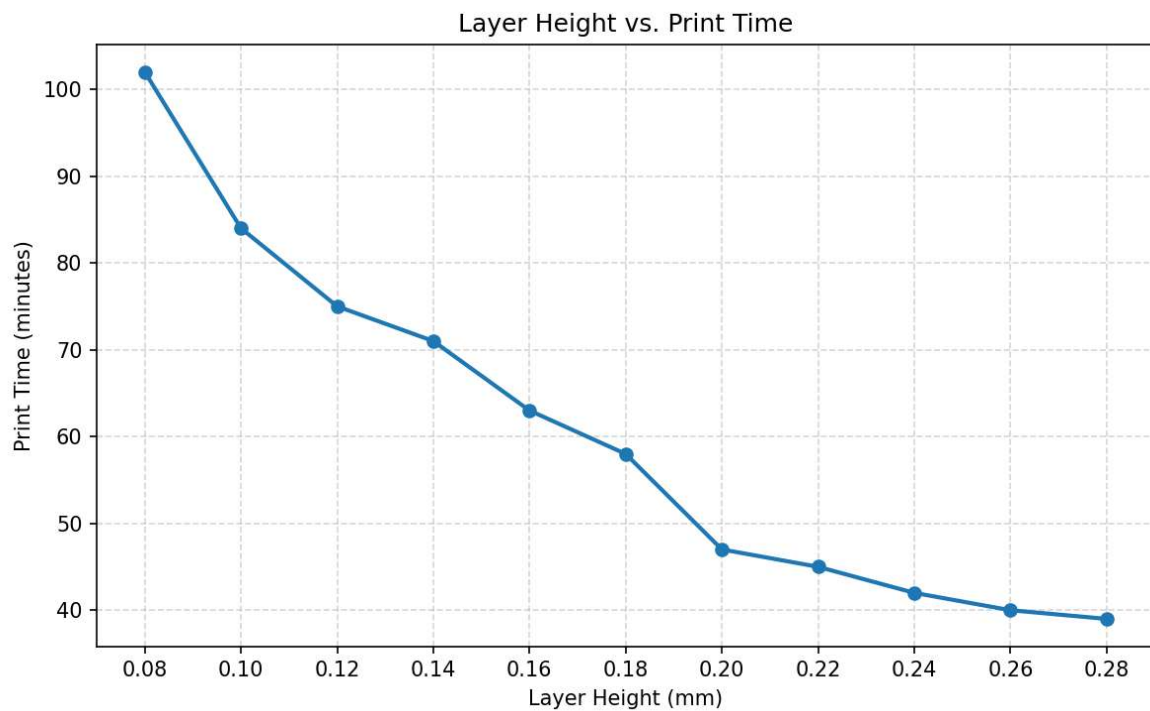
Figure 10: Part Sliced At Standard Settings In Bambu Studio

Using Bambu Studio we were able to generate a table of print times as a function of layer height from 0.08mm to 0.28mm in equal increments.

Layer Height (mm):	Print Time (minutes):
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0.08	102
0.10	84
0.12	75
0.14	71
0.16	63
0.18	58
0.20	47
0.22	45
0.24	42
0.26	40
0.28	39

Note: These values were generated by taking Bambu Studio's 'Standard' settings and varying only layer height. Initial layer height (which is its own setting) was kept at 0.20mm as layer height (for all other layers) was varied.



We can build a more formally defined model for how layer height relates to print time by relating the print time to the number of layers in a given print as a result of the layer height and determining average layer time which should be approximately equal or at least cluster around the same value as layer height varies.

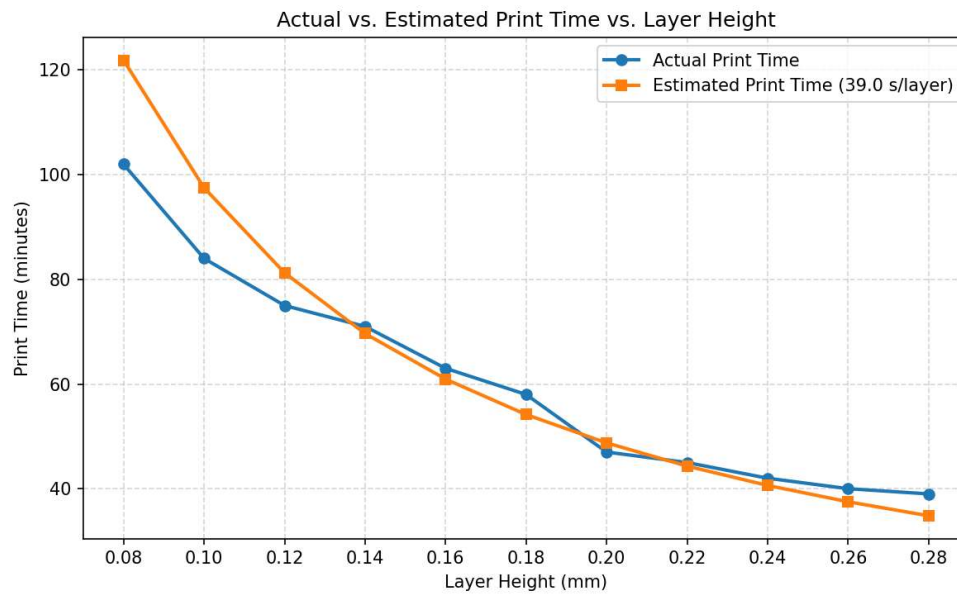
We get the number of layers from the readout of our slicer after inputting each of the above layer heights, then we divide the time taken per print by the number of layers in that print to determine that print's average layer height.

As seen in the above graph, the average layer time of each tested print sits between 33.0 seconds and 43.0 seconds with the mean of the average layer times recorded being 39.0 seconds. We can use this to estimate the part's total print time from its layer height.

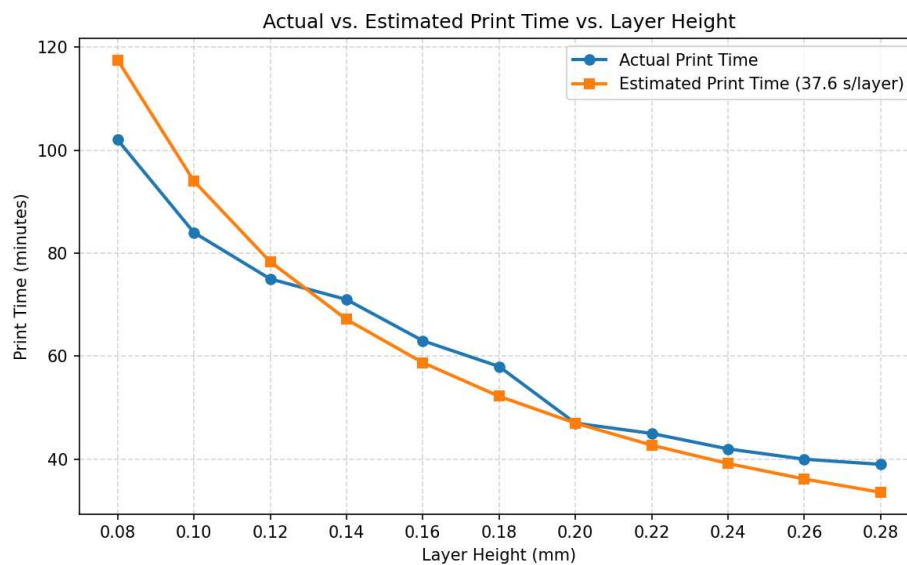
We can estimate the number of layers that the printer will lay down by taking the maximum height of our assembly as it is oriented on the build plate (15.0mm) and dividing by the layer height. This quotient is not a definitive formula for the number of layers used in the printing process as initial layer height is being held constant at 0.2mm, but it is an adequate estimate for our purposes.

$$\text{Print Time} \approx \frac{\text{Part Height}}{\text{Layer Height}} \cdot \text{Average Layer Time}$$

By multiplying our estimate for the number of layers in our print by the average time per layer we can estimate the time it will take for our print to complete.



The above model for estimated print time uses an average layer time taken by finding the mean of various print times determined by the Bambu Studio slicer. An adequate model can be instead found by substituting the mean of the average layer times with the average layer time of the printing process performed when layer height is centered in the range of values we wish to find estimates for.



In the above graph our model for estimated print time as a function of layer height uses 37.6 seconds per layer as the average layer time, which was found by taking the average layer time of our print when a layer height of 0.20 was used.

$$\text{Print Time} \approx \frac{\text{Part Height}}{\text{Layer Height}} \cdot \text{Representative Layer Time}$$

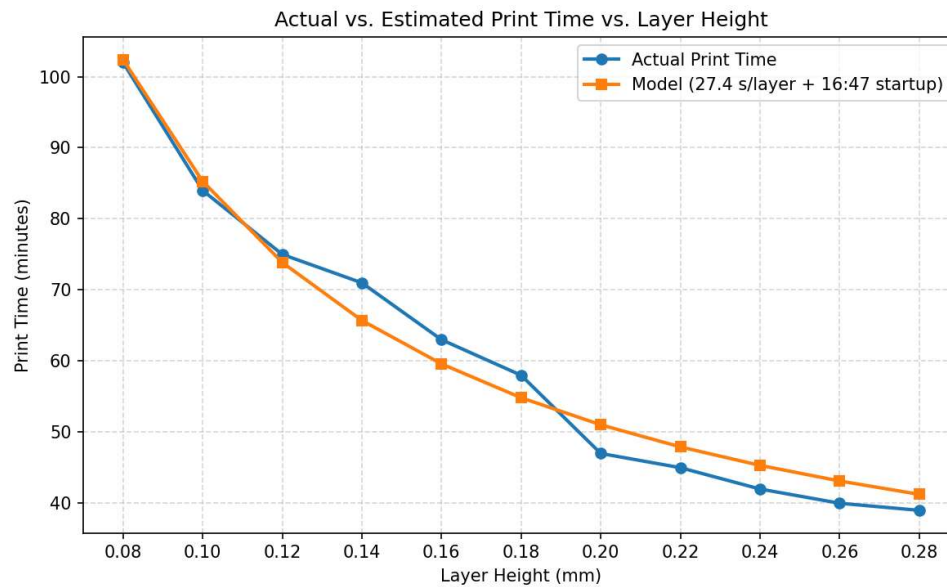
The above formula does not account for the printer's initial start up time, which can be assumed to be constant as layer height varies and all other settings are kept constant (this was verified by checking startup times in Bambu Studio across various layer heights while other settings were kept constant).

The data in the Slicing Result window of Bambu Studio indicates that the time it takes for the printer to carry out the startup process (purging existing filament, cleaning the nozzle, and adequately heating the print bed) is 9 minutes and 39 seconds. After that the first layer (which takes the same amount of time as layer height changes) always prints in 7 minutes and 8 seconds. The total startup time is the sum of these two times, 16 minutes and 47 seconds.

The inclusion of a minimum amount of time a print will need that is constant as layer height varies will be reflected in our mathematical model through the addition of a constant.

$$\text{Print Time} \approx \left(\frac{\text{Part Height}}{\text{Layer Height}} \cdot \text{Representative Layer Time} \right) + \text{Startup Time}$$

Using the above function we can refine our mathematical model even further by using least squares to find a value for our Representative Layer Time that makes our model fit the given data as closely as possible. After running a least squares regression we get that the optimal value for our Representative Layer Time is 27.4 seconds per layer. Previously we had included the time taken for the first layer and the startup time in our calculation of average layer time, how that startup time is being properly accounted for in our model it makes sense that our estimates for layer time shrink.



In totality the form of our mathematical model for print time as a function of layer height is:

$$\text{Print Time} \approx \left(\frac{\text{Part Height}}{\text{Layer Height}} \cdot \text{Representative Layer Time} \right) + \text{Startup Time}$$

Where:

- Part Height = 15.0mm
- Representative Layer Time = 27.4 seconds per layer
- Startup Time = 16.78 minutes

This model's efficacy is dependent on Startup Time being constant across multiple prints. If the printer sees continuous use then the print bed may already be warm upon the start of the startup process, meaning the total print will take less time than the model would currently predict.

More generally this model's efficacy is dependent on the assumptions used in its deviation:

- That what is referred to as 'Representative Layer Time' is close to the average value of the average layer time, and also that there's little variance in layer time for all the layers of a single print.
- That no other slicer settings vary across the model's use except for layer height.

To make a more precise estimate we would need information on the per-layer process steps of the print. The modeling approach used above looks at layer print times in aggregate, if the model could account for things like, the amount of time that the toolhead spent at steady

state, or at what point in the print the toolhead stops switching from printing one part to another thereby reducing time spent traveling, this would allow for the creation of a more precise method of estimating print time.

Conclusion

This project showed that both FFF and SLA can be used to manufacture the gear fidget toy, but each process introduces different tradeoffs in dimensional accuracy, surface quality, and assembly performance. FFF produced a more forgiving snap-fit interaction and required less post-processing, but it showed greater corner rounding defects near the base of the teeth and visible first-layer surface defects. SLA produced sharper geometry and better surface detail on the teeth and gear surfaces, but support marks, brittle behavior during post-processing, and tighter-than-intended fit in the peg-hole connection reduced its functional robustness.

To improve the design for additive manufacturing, the mating peg-hole geometry should be modified with process-specific tolerance offsets rather than using one nominal CAD fit for both methods. For FFF, improvements should focus on first-layer adhesion, extrusion calibration, and possibly a finer layer height to reduce the aforementioned tooth rounding issue. For SLA, improvements should focus on support placement, more conservative handling of delicate features, and dimensional compensation for shrinkage and tight-fit interfaces. If cost were not a concern, SLA would be the better process for producing this part when visual quality and fine feature definition are the highest priorities. However, for a product expected to be handled repeatedly by a customer, FFF may currently be the more viable option because it produces a more durable and functional assembly with fewer failures during handling.

Picture of Parts

A score (0/5/10/15) will also be assigned to indicate the quality represented by your finished parts; this will be judged relative to the difficulty; if you execute well, you can expect full credit.





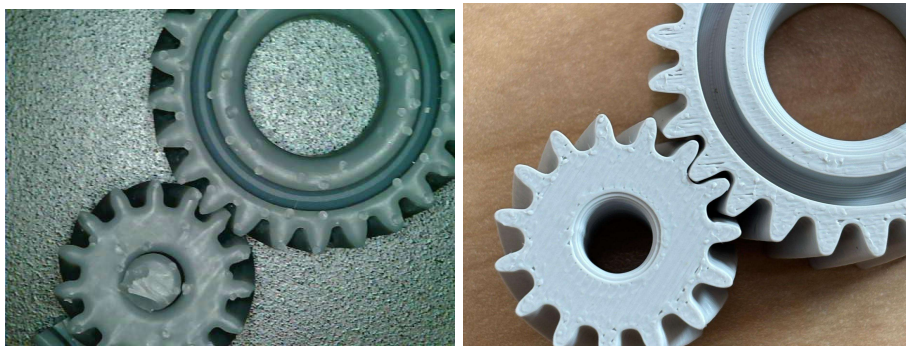








Note: First rough assembly before post processing (above)





Deliverables

https://drive.google.com/drive/folders/1JvO4Tol-85N9IR1mV2OMGNZAi7Kz3l9G?usp=drive_link

